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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

PROJECT TITLE

Submitted by

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Under the Guidance of

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DECLARATION

I hereby declare that the project entitled "**Personal Expense Tracking and Monthly Budget Management System"** submitted in partial fulfillment of the requirements for the course **BSc Information Technology** is a **record of my original work** carried out under the guidance of **Mugilan R** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

harishmitha G

CERTIFICATE

This is to certify that the project entitled **"Personal Expense Tracking and Monthly Budget Management System"** is a **bona fide work** carried out by **Harishmitha.G, B.Sc. Technology - 'B'**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming** .

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The **Personal Expense Tracking and Monthly Budget Management System** is a software application developed to help users manage their daily expenses and monthly budget in an easy and organized way. In daily life, many people spend money on different needs such as food, travel, shopping, bills, and entertainment. If these expenses are not recorded properly, it becomes difficult to know where the money is being spent.

This system helps the user to enter income, record expenses under different categories, and calculate the total spending for a month. It also compares the user's expenses with the monthly budget and shows whether the user is within budget or has exceeded the limit.

The system is developed using **Core Java** concepts. It provides a simple and user-friendly method for maintaining personal financial records. This project improves financial awareness, reduces manual calculation errors, saves time, and helps users plan their monthly expenses better.

Overall, the **Personal Expense Tracking and Monthly Budget Management System** is useful for managing personal finance in an effective and efficient way.

1. INTRODUCTION

In the hospitality industry, efficient billing systems play a crucial role in managing restaurant operations and ensuring customer satisfaction. Traditionally, many small restaurants and hotels rely on manual billing methods, where staff calculate the total bill using handwritten notes or simple calculators. This manual process is time-consuming and can often lead to calculation errors, delays in service, and difficulty in maintaining proper billing records.

In today's world, managing money properly is very important in everyone's life. People earn money and spend it on many daily needs such as food, travel, shopping, electricity bills, mobile recharge, education, medical expenses, and entertainment. If a person does not keep track of these expenses, it becomes very difficult to know how much money is spent and how much money is left at the end of the month.

Many people still write their expenses in notebooks or try to remember their spending mentally. This manual method is not always accurate and may lead to mistakes. Sometimes people spend more than their monthly budget without realizing it. This creates financial problems and makes saving money difficult. So, there is a need for a simple system that can record expenses properly and help in budget management.

The **Personal Expense Tracking and Monthly Budget Management System** is developed to solve this problem. This system helps the user to enter monthly income, set a monthly budget, and record expenses under different categories. After entering the values, the system automatically calculates the total expenses and remaining balance. It also checks whether the user is staying within the budget or exceeding the budget.

This project is developed using **Core Java** programming concepts. It is designed as a simple and user-friendly application so that any user can easily understand and use it. The system reduces manual work, improves calculation accuracy, and helps the user maintain financial discipline.

The main purpose of this project is to make personal finance management easy, clear, and efficient. It also shows how Java programming can be used to develop useful real-life applications. This system is especially helpful for students, employees, and families who want to control their monthly spending and manage their budget in a better way.

1.2 Overview

The main objective of the **Personal Expense Tracking and Monthly Budget Management System** is to develop a simple and efficient application using **Core Java** for managing personal income, daily expenses, and monthly budget in an organized manner. This system is designed to help users record their spending details and understand how their money is being used every month.

Another important objective of this project is to reduce the difficulties faced in manual expense tracking. When expenses are recorded manually in notebooks or calculated mentally, there is a high chance of forgetting entries, making calculation mistakes, and losing track of the budget. This system helps to avoid such problems by automatically calculating total expenses and remaining balance.

The project also aims to help users build better financial habits. By entering expenses under different categories such as food, travel, shopping, bills, and entertainment, users can clearly identify where more money is being spent. This makes it easier to control unnecessary expenses and maintain proper budget discipline.

1.3 Features

To develop a Java-based application for personal expense tracking.

- To store monthly income and budget details.
- To record expenses under different categories.
- To calculate the total expenses automatically.
- To find the remaining balance after expenses.
- To compare expenses with the monthly budget.
- To display whether the user is within budget or has exceeded the budget.
- To reduce manual calculation errors.
- To provide a simple and user-friendly system for personal finance management.
- To demonstrate Core Java concepts such as variables, methods, conditional statements, loops, and input handling using Scanner.

Overall, the objective of this project is not only to develop a working Java application, but also to create a useful real-life system that helps users manage money effectively and improve monthly financial planning

2. PROGRAM ANALYSIS

Program analysis helps to understand the requirements and working process of the **Personal Expense Tracking and Monthly Budget Management System**. In this system, the user enters income, monthly budget, and expense details. The system processes the data and provides the financial summary.

The system uses Core Java concepts such as variables, loops, conditional statements, methods, and input handling. This reduces manual work and improves accuracy.

2.1 Problem Definition

In daily life, many people face problems in managing their personal expenses. Some common problems are:

- Expenses are not recorded regularly.
- Manual calculation may lead to errors.
- It is difficult to identify where money is being spent.
- Monthly budget planning becomes difficult.
- Overspending may happen without proper tracking.
- There is no clear report of remaining balance.

Therefore, a simple computerized system is required to record expenses and compare them with the monthly budget. The **Personal Expense Tracking and Monthly Budget Management System** helps to solve these problems by providing an organized and accurate solution

2.2 Project Overview

The **Personal Expense Tracking and Monthly Budget Management System** is a Java-based application developed to manage personal financial records in a simple way.

The system allows the user to:

- Enter monthly income
- Set a monthly budget
- Add daily expenses

- Calculate total expenses
- Find remaining balance
- Check budget status

This system helps users reduce unnecessary spending and improve money management. It is easy to use and suitable for basic personal finance management.

This project also helps the user maintain financial discipline by giving a clear view of monthly spending and available balance

2.3 Scopes and Limitations

Scope

The scope of the system includes:

- Storing monthly income and budget details
- Recording daily expenses
- Categorizing expenses
- Calculating total expenses automatically
- Showing remaining balance
- Checking whether the budget is exceeded
- Helping users manage personal finances better

Limitations

The limitations of the system include:

- The system is designed only for basic personal expense tracking.
- It does not support online banking integration.
- It does not store data permanently in a database.
- It may handle only one user at a time.
- It does not include graphical charts or advanced reports.
- All values must be entered manually by the user

3. SYSTEM DESIGN

The **Personal Expense Tracking and Monthly Budget Management System** is designed to help users manage their monthly income, expenses, and budget in a simple way. The system accepts expense details from the user, calculates total expenses, checks the remaining balance, and shows the budget status.

The system is divided into the following modules:

3.1 Modules

The **Personal Expense Tracking and Monthly Budget Management System** is divided into different modules to make the working process simple, clear, and easy to understand. Each module performs a specific function in the system. These modules work together to record income, store expenses, calculate totals, check budget status, and display the final result

1. Icome and Budget module

This module is used to enter and store the user's monthly income and budget details. It helps the system set the financial limit for expense calculation and budget checking.

Functions:

- Accept monthly income
- Accept monthly budget amount
- Store income and budget details for further processing

Example:

- Monthly Income: 25000
- Monthly Budget: 20000

This module forms the base for the entire system because it provides the financial details required for expense management and budget comparison

2. Expense Entry module

This module is used to enter expenses under different categories such as food, travel, shopping, bills, and entertainment.

Functions:

- Calculate total expenses
- Calculate remaining balance
- Compare total expenses with income and budget

Example:

- Total Expenses = Food + Travel + Shopping + Bills + Entertainment
- Remaining Balance = Income - Total Expenses

This module reduces manual calculation errors and helps the user get quick and accurate results

3. Calculation Module

This module performs all calculations related to expenses and budget.

Functions:

- Calculate total expenses
- Calculate remaining balance
- Compare expenses with budget

4. Result Display Module

This module generates the final bill for the customer.

This module displays the complete expense summary to the user.

Functions:

- Display user name
- Display income and budget
- Display all expense details
- Display total expenses
- Display remaining balance
- Display budget status

5. Exit Module

This module is used to end the program safely.

Function:

- Exit the application

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from identifying the problem to designing, implementing, testing, and documenting the system.

For the **Personal Expense Tracking and Monthly Budget Management System**, a structured development method was followed to make sure the system works properly and meets user needs.

The development stages are:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem of managing personal expenses manually is studied. The main issues identified are:

- Lack of proper expense recording
- Difficulty in tracking monthly spending
- Manual errors in total calculation
- Poor budget planning
- Difficulty in knowing remaining balance

Based on these issues, the system requirements were defined

4.2. System Design

In this stage, the structure of the system is designed before coding begins. The system is divided into modules such as:

- Income and Budget Module
- Expense Entry Module
- Expense Calculation Module
- Budget Status Module
- Result Display Module
- Exit Module

The flow of the program is planned in a simple way so that users can easily interact with the system.

4.3. Program Development

After system design, the program is developed using Java programming language.

The following steps were followed:

- Creation of variables and methods
- Accepting user input using Scanner
- Writing logic for expense and budget calculation
- Applying conditional statements for budget status
- Displaying final result clearly

Core Java concepts such as classes, methods, loops, variables, and conditions were used.

4.4. Testing and Validation

Testing is done to make sure the system works correctly.

Example Test Cases:

Test Case	Expected Result
Enter income and budget	Values stored correctly
Enter expense details	Expenses accepted properly
Calculate total expenses	Correct total displayed
Check remaining balance	Correct balance shown
Budget comparison	Proper status displayed
Invalid input	Error handled properly

This phase ensures that the program produces correct results

4.5. Result and Documentation

After successful testing, the system gives the correct output for expense tracking and monthly budget calculation. The application helps users manage spending and plan finances properly.

All phases of the project such as analysis, design, development, and testing are documented clearly for understanding the project.

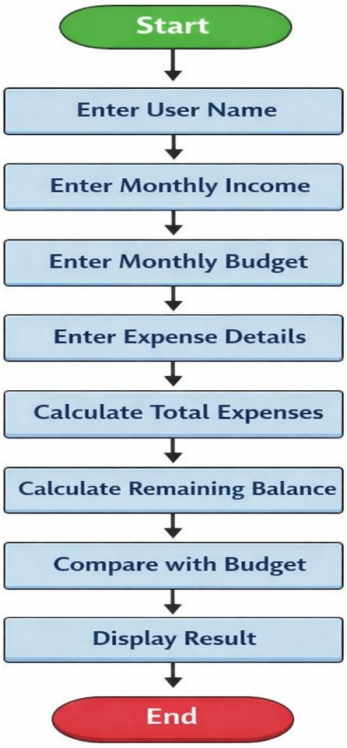
4.6. Algorithm

Algorithm for Personal Expense Tracking and Monthly Budget Management System:

1. Start the program.
2. Enter the user name.

3. Enter monthly income.
4. Enter monthly budget.
5. Enter expenses under different categories.
6. Calculate total expenses.
7. Calculate remaining balance.
8. Compare total expenses with monthly budget.
9. Display financial summary.
10. Show budget status.
11. End the program

4.7. Flowchart



5. SOURCE CODE

```
import java.util.Scanner;

public class PersonalExpenseTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String name;
        double income, budget;
        double food, travel, shopping, bills, entertainment;
        double totalExpense, balance;

        System.out.println("----- Personal Expense Tracking and Monthly Budget Management
System -----");

        System.out.print("Enter User Name: ");
        name = sc.nextLine();

        System.out.print("Enter Monthly Income: ");
        income = sc.nextDouble();

        System.out.print("Enter Monthly Budget: ");
```

```
budget = sc.nextDouble();

System.out.print("Enter Food Expense: ");
food = sc.nextDouble();

System.out.print("Enter Travel Expense: ");
travel = sc.nextDouble();

System.out.print("Enter Shopping Expense: ");
shopping = sc.nextDouble();

System.out.print("Enter Bills Expense: ");
bills = sc.nextDouble();

System.out.print("Enter Entertainment Expense: ");
entertainment = sc.nextDouble();

totalExpense = food + travel + shopping + bills + entertainment;
balance = income - totalExpense;

System.out.println("\n------ MONTHLY EXPENSE REPORT ------");
System.out.println("User Name: " + name);
System.out.println("Monthly Income: " + income);
System.out.println("Monthly Budget: " + budget);

System.out.println("\nExpense Details:");
System.out.println("Food: " + food);
System.out.println("Travel: " + travel);
System.out.println("Shopping: " + shopping);
```

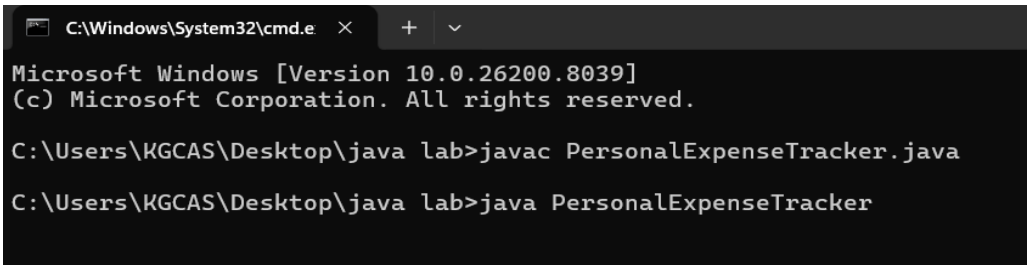
```
        System.out.println("Bills: " + bills);
        System.out.println("Entertainment: " + entertainment);

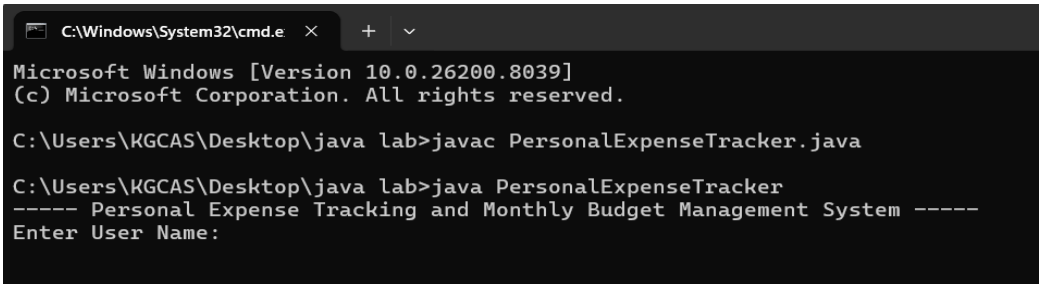
        System.out.println("\nTotal Expenses: " + totalExpense);
        System.out.println("Remaining Balance: " + balance);

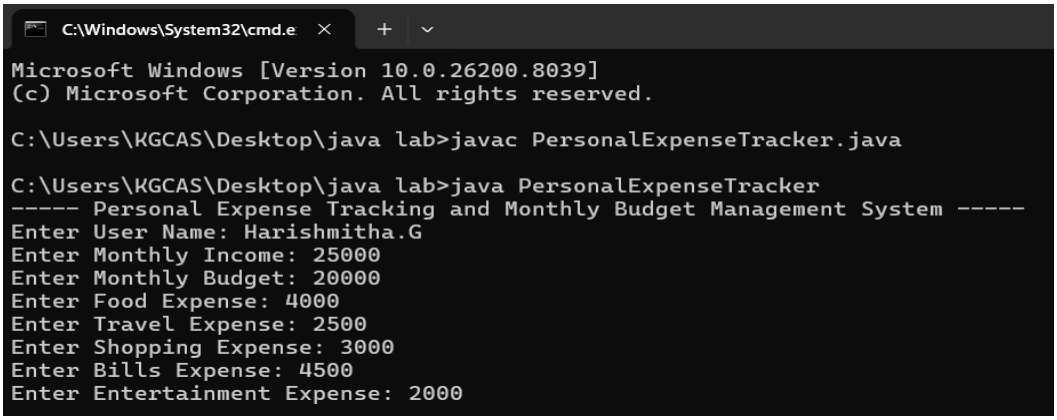
        if (totalExpense <= budget) {
            System.out.println("Budget Status: Within Budget");
        } else {
            System.out.println("Budget Status: Budget Exceeded");
        }

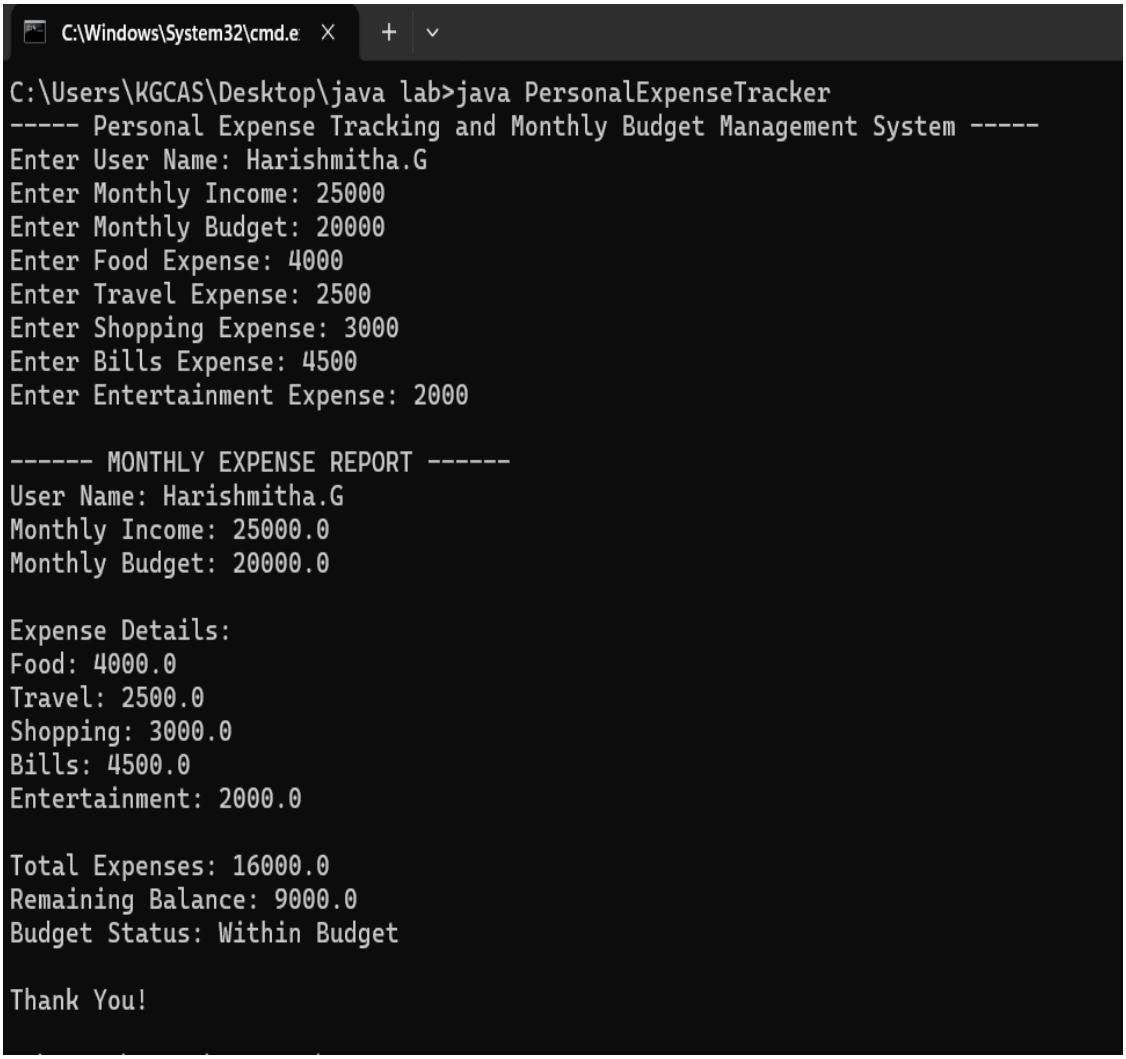
        System.out.println("\nThank You!");
        sc.close();
    }
}
```

6. INPUT AND OUTPUT SCREENS








C:\Windows\System32\cmd.e
C:\Users\KGCAS\Desktop\java lab>java PersonalExpenseTracker
----- Personal Expense Tracking and Monthly Budget Management System -----
Enter User Name: Harishmitha.G
Enter Monthly Income: 25000
Enter Monthly Budget: 20000
Enter Food Expense: 4000
Enter Travel Expense: 2500
Enter Shopping Expense: 3000
Enter Bills Expense: 4500
Enter Entertainment Expense: 2000
------ MONTHLY EXPENSE REPORT ------
User Name: Harishmitha.G
Monthly Income: 25000.0
Monthly Budget: 20000.0
Expense Details:
Food: 4000.0
Travel: 2500.0
Shopping: 3000.0
Bills: 4500.0
Entertainment: 2000.0
Total Expenses: 16000.0
Remaining Balance: 9000.0
Budget Status: Within Budget
Thank You!

7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Personal Expense Tracking and Monthly Budget Management System** was developed to simplify the process of managing personal income and expenses. The system allows the user to enter income, budget, and expense details. It automatically calculates total expenses and remaining balance.

During the development of this project, Core Java concepts such as variables, methods, Scanner class, conditional statements, and simple calculations were used. These concepts helped in building a simple financial management application.

The system was tested with different input values to ensure that the calculations were correct. The results show that the system works properly and reduces manual effort in personal budget tracking

7.2 Conclusion

The **Personal Expense Tracking and Monthly Budget Management System** successfully demonstrates the practical use of Core Java programming concepts in developing a real-life financial application.

The system helps users track their expenses, control their spending, and manage their monthly budget effectively. It reduces calculation errors and provides a simple and organized method for personal finance management.

In future, the system can be improved by adding database support, graphical reports, category-wise analysis, and a graphical user interface.

8. REFERENCES

The following resources were used during the development of this project:

Book

Herbert Schildt – *Java: The Complete Reference*

Websites

1. Oracle Java Documentation – https://docs.oracle.com/en/java/
2. GeeksforGeeks Java Tutorial https://www.Geeksforgeeks .org/Java/Java/
3. W3Schools Java Tutorial – https://www.w3schools.com/ java/
4. Javatpoint Java Tutorial – https://www.tpointtech.com /java-tutorial